

Envelope Equations

USPAS - Beam Physics with Intense Space Charge

Austin Hoover Steven Lund Daniel Winklehner John Barnard

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Paraxial ray equation

Envelope equations for axially symmetric beams

Envelope equations for elliptically symmetric beams

Envelope equations: reduced-order characterization of beam dynamics in terms of second-order moments

[Image]

Road map

- Start from single particle equation of motion: $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$.
- Make use of:
 - Paraxial (near-axis) approximation
 - Conservation of angular momentum
 - Axisymmetry of external forces
- Arrive at single-particle paraxial ray equation.
- Take statistical averages to obtain moment/envelope equations.

Paraxial ray equation

Equation of motion in cylindrical coordinates

Equation of motion for particle of mass m and charge q in electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$:

$$\frac{d\mathbf{p}}{dt} = \frac{d(\gamma m \dot{\mathbf{x}})}{dt} = q(\mathbf{E} + \dot{\mathbf{x}} \times \mathbf{B}), \quad (1)$$

Switch to cylindrical coordinates (r, θ, z) :

$$\begin{aligned} \mathbf{p} &= p_x \hat{x} + p_y \hat{y} + p_z \hat{z} \\ &= p_r \hat{r} + p_\theta^* \hat{\theta} + p_z \hat{z}. \end{aligned} \quad (2)$$

The components p_r and p_θ^* are given by

$$\begin{aligned} p_r &= \gamma m \dot{r}, \\ p_\theta^* &= \gamma m r \dot{\theta}. \end{aligned} \quad (3)$$

The symbol p_θ^* is used to reserve p_θ for the *canonical* angular momentum, defined later.

In cylindrical coordinates, the unit vectors depend on the particle position and thus change with time:

$$\begin{aligned}\dot{\hat{r}} &\equiv d\hat{r}/dt = \dot{\theta}\hat{\theta}, \\ \dot{\hat{\theta}} &\equiv d\hat{\theta}/dt = -\dot{\theta}\hat{r}.\end{aligned}\tag{4}$$

Thus, the time derivative of the momentum vector in Eq. (2) is:

$$\frac{d\mathbf{p}}{dt} = \frac{d}{dt} (p_r\hat{r} + p_\theta^*\hat{\theta} + p_z\hat{z})\tag{5}$$

$$= \dot{p}_r\hat{r} + p_r\dot{\hat{r}} + \dot{p}_\theta\hat{\theta} + p_\theta^*\dot{\hat{\theta}} + \dot{p}_z\hat{z}\tag{6}$$

$$= (\dot{p}_r - p_\theta^*\dot{\theta})\hat{r} + (\dot{p}_\theta + p_r\dot{\theta})\hat{\theta} + \dot{p}_z\hat{z}\tag{7}$$

$$= \left[\frac{d(\gamma m \dot{r})}{dt} - \gamma m r \dot{\theta}^2 \right] \hat{r} + \left[\frac{d(\gamma m r \dot{\theta})}{dt} + \gamma m \dot{r} \dot{\theta} \right] \hat{\theta} + \left[\frac{d(\gamma m \dot{z})}{dt} \right] \hat{z}.\tag{8}$$

$$= \left[\frac{d(\gamma m \dot{r})}{dt} - \gamma m r \dot{\theta}^2 \right] \hat{r} + \left[\frac{1}{r} \frac{d}{dt} (\gamma m r^2 \dot{\theta}) \right] \hat{\theta} + \left[\frac{d(\gamma m \dot{z})}{dt} \right] \hat{z}.\tag{9}$$

This takes care of the left side of the Lorentz force equation (Eq. (1)).

The right side of Eq. (1) requires the cylindrical field components:

$$\begin{aligned}\mathbf{E} &= E_r \hat{r} + E_\theta \hat{\theta} + E_z \hat{z}, \\ \mathbf{B} &= B_r \hat{r} + B_\theta \hat{\theta} + B_z \hat{z}.\end{aligned}\tag{10}$$

Evaluating the cross product $\dot{\mathbf{x}} \times \mathbf{B}$,

$$\dot{\mathbf{x}} \times \mathbf{B} = \begin{vmatrix} \hat{r} & \hat{\theta} & \hat{z} \\ \dot{r} & r\dot{\theta} & \dot{z} \\ B_r & B_\theta & B_z \end{vmatrix},\tag{11}$$

leads to

$$\frac{d(\gamma m \dot{r})}{dt} - \gamma m r \dot{\theta}^2 = q (E_r + r \dot{\theta} B_z - \dot{z} B_\theta),\tag{12}$$

$$\frac{1}{r} \frac{d(\gamma m r^2 \dot{\theta})}{dt} = q (E_\theta + \dot{z} B_r - \dot{r} B_z),\tag{13}$$

$$\frac{d(\gamma m \dot{z})}{dt} = q (E_z + \dot{r} B_\theta - r \dot{\theta} B_r)\tag{14}$$

From fields to potentials

Write the fields in terms of the scalar potential $\phi(\mathbf{x})$ and vector potential $\mathbf{A}(\mathbf{x})$:

$$\begin{aligned}\mathbf{E} &= -\nabla\phi - \frac{\partial\mathbf{A}}{\partial t}, \\ \mathbf{B} &= \nabla \times \mathbf{A}.\end{aligned}\tag{15}$$

The gradient $\nabla\phi$ in cylindrical coordinates is:

$$\nabla\phi = \left[\frac{\partial\phi}{\partial r}\right]\hat{r} + \left[\frac{1}{r}\frac{\partial\phi}{\partial\theta}\right]\hat{\theta} + \left[\frac{\partial\phi}{\partial z}\right]\hat{z}.\tag{16}$$

And the curl $\nabla \times \mathbf{A}$ is:

$$\nabla \times \mathbf{A} = \left[\frac{1}{r}\frac{\partial A_z}{\partial\theta} - \frac{\partial A_\theta}{\partial z}\right]\hat{r} + \left[\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r}\right]\hat{\theta} + \frac{1}{r}\left[\frac{\partial(rA_\theta)}{\partial r} - \frac{\partial A_r}{\partial\theta}\right]\hat{z}.\tag{17}$$

Axial symmetry means all partial derivatives with respect to θ are zero in Eq. (16) and Eq. (17). This gives the following expressions for the fields in terms of the potentials in cylindrical coordinates:

$$\begin{aligned}\mathbf{E} &= \left[-\frac{\partial\phi}{\partial r} - \frac{\partial A_r}{\partial t} \right] \hat{r} + \left[-\frac{\partial A_\theta}{\partial t} \right] \hat{\theta} + \left[-\frac{\partial\phi}{\partial z} - \frac{\partial A_z}{\partial t} \right] \hat{z}, \\ \mathbf{B} &= \left[-\frac{\partial A_\theta}{\partial r} \right] \hat{r} + \left[\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \right] \hat{\theta} + \left[\frac{1}{r} \frac{\partial(rA_\theta)}{\partial r} \right] \hat{z}.\end{aligned}\tag{18}$$

Conservation of angular momentum

Use the potentials to rewrite the angular equation of motion (Eq. (13)):

$$\begin{aligned}\frac{d}{dt}(\gamma m r^2 \dot{\theta}) &= q r (E_{\theta} + \dot{z} B_r - \dot{r} B_z) \\&= q r \left(\left[-\frac{\partial A_{\theta}}{\partial t} \right] + \dot{z} \left[-\frac{\partial A_{\theta}}{\partial z} \right] - \dot{r} \left[\frac{1}{r} \frac{\partial(r A_{\theta})}{\partial r} \right] \right) \\&= -q \left(\frac{\partial(r A_{\theta})}{\partial t} + \dot{z} \frac{\partial(r A_{\theta})}{\partial z} + \dot{r} \frac{\partial(r A_{\theta})}{\partial r} \right) \\&= -q \left[\frac{\partial}{\partial t} + \dot{z} \frac{\partial}{\partial z} + \dot{r} \frac{\partial}{\partial r} \right] (r A_{\theta}) \\&= -q \left[\frac{\partial}{\partial t} + \dot{\mathbf{x}} \cdot \nabla \right] (r A_{\theta}) \\&= -\frac{d(q r A_{\theta})}{dt}.\end{aligned}\tag{19}$$

Eq. (19) can then be written as:

$$\frac{d}{dt} (\gamma m r^2 \dot{\theta} + q r A_{\theta}) = 0. \quad (20)$$

This motivates the definition of the *canonical angular momentum*:

$$p_{\theta} \equiv \gamma m r^2 \dot{\theta} + q r A_{\theta}, \quad (21)$$

such that $dp_{\theta}/dt = 0$. We may also write p_{θ} in terms of the magnetic flux $\psi(r)$ though a circle of radius r [Figure]:

$$\psi(r) = \int \mathbf{B} \cdot d\Omega = \int \nabla \times \mathbf{A} \cdot d\Omega = \oint \mathbf{A} \cdot d\mathbf{l} = 2\pi r A_{\theta}. \quad (22)$$

Therefore, Eq. (21) can be written

$$p_{\theta} \equiv \gamma m r^2 \dot{\theta} + \frac{q\psi(r)}{2\pi}. \quad (23)$$

Series expansion of axially symmetric fields

Consider external fields $\mathbf{E}$ and $\mathbf{B}$ with azimuthal symmetry ($\partial/\partial\theta = 0$). In the absence of charge and current densities,

$$\begin{aligned}\nabla \cdot \mathbf{E} &= 0, \\ \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= 0, \\ \nabla \times \mathbf{B} &= 0.\end{aligned}\tag{24}$$

The vanishing curl means we can express the fields as

$$\begin{aligned}\mathbf{E} &= -\nabla\phi, \\ \mathbf{B} &= -\nabla\phi_m,\end{aligned}\tag{25}$$

for scalar potentials ϕ and ϕ_m . Additionally, the vanishing divergence leads to the Laplace equation for the scalar potential:

$$\nabla^2\phi = 0.\tag{26}$$

In cylindrical coordinates, the Laplace equation for the potential is:

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{\partial^2 \phi}{\partial z^2} = 0. \quad (27)$$

The azimuthal symmetry will lead to a simplified form of the scalar potential in terms of its on-axis derivatives along the z axis. First, expand the potential $\phi(r, z)$ as a power series,

$$\phi(r, z) = \sum_{n=0}^{\infty} h_{2n}(z) r^{2n}, \quad (28)$$

where the $h_{2n}(z)$ are functions of z . The on-axis potential is given by $h_0(z) = \phi(0, z)$.

The rotational symmetry eliminates all odd powers of r in the series.

At the origin ($r = 0$), we must have $E_r = B_r = 0$. If we allowed the $n = 1$ term $h_1(z)r$, we would have $E_r(0, z) = -\partial\phi(r, z)/\partial r|_{r=0} = h_1(z) \neq 0$.

The derivatives in the Laplace equation are:

$$\begin{aligned}\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) &= \sum_{n=1}^{\infty} (2n)^2 h_{2n}(z) r^{2n-2}, \\ \frac{\partial^2 \phi}{\partial z^2} &= \sum_{n=0}^{\infty} h_{2n}''(z) r^{2n},\end{aligned}\tag{29}$$

where $h'(z) = \partial h(z) / \partial z$.

Thus, $\nabla^2\phi = 0$ becomes:

$$\begin{aligned}
\sum_{n=1}^{\infty} (2n)^2 h_{2n}(z) r^{2n-2} + \sum_{n=0}^{\infty} h_{2n}''(z) r^{2n} &= 0 \\
\sum_{n=0}^{\infty} (2n+2)^2 h_{2n+2}(z) r^{2n} + \sum_{n=0}^{\infty} h_{2n}''(z) r^{2n} &= 0 \\
\sum_{n=0}^{\infty} [(2n+2)^2 h_{2n+2}(z) + h_{2n}''(z)] &= 0
\end{aligned} \tag{30}$$

The result is a recurrence relation,

$$h_{2n+2}(z) = -\frac{h_{2n}''(z)}{(2n+2)^2}, \tag{31}$$

with which we can write all terms as derivatives of $h_0(z)$ with respect to z :

$$\begin{aligned}
h_2(z) &= -\frac{1}{4} \frac{\partial^2 h_0(z)}{\partial z^2}, \\
h_4(z) &= +\frac{1}{64} \frac{\partial^4 h_0(z)}{\partial z^4}, \\
&\vdots
\end{aligned} \tag{32}$$

We may thus write the potential at all points in space using only the z -derivatives of the on-axis potential $\phi(0, z)$:

$$\phi(r, z) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} \frac{\partial^{2n} \phi(0, z)}{\partial z^{2n}} \left(\frac{r}{2}\right)^{2n}. \quad (33)$$

We now express the fields in terms of the expanded scalar potentials. Write the magnetic field as $\mathbf{B} = B_r \hat{r} + B_z \hat{z}$. Using the series expansion of $\phi_m(r, z)$, we find for the z component of the field:

$$\begin{aligned}
B_z(r, z) &= -\frac{\partial \phi_m}{\partial z} \\
&= -\frac{\partial}{\partial z} \left[\sum_{n=0}^{\infty} \frac{(-1)^{2n}}{(n!)^2} \frac{\partial^{2n} \phi_m(0, z)}{\partial z^{2n}} \left(\frac{r}{2}\right)^2 \right] \\
&= -\sum_{n=0}^{\infty} \frac{(-1)^{2n}}{(n!)^2} \frac{\partial^{2n+1} \phi_m(0, z)}{\partial z^{2n+1}} \left(\frac{r}{2}\right)^2.
\end{aligned} \tag{34}$$

Define $B(z)$ as the on-axis value of $B_z(z)$:

$$B(z) \equiv B_z(0, z) = -\frac{\partial \phi_m(0, z)}{\partial z}. \tag{35}$$

Inserting Eq. (35) into Eq. (34) gives:

$$\begin{aligned}
B_z(r, z) &= \sum_{n=0}^{\infty} \frac{(-1)^{2n}}{(n!)^2} \frac{\partial^{2n}}{\partial z^{2n}} \left[-\frac{\partial \phi_m(0, z)}{\partial z} \right] \left(\frac{r}{2} \right)^2 \\
B_z(r, z) &= \sum_{n=0}^{\infty} \frac{(-1)^{2n}}{(n!)^2} \frac{\partial^{2n} B(z)}{\partial z^{2n}} \left(\frac{r}{2} \right)^2.
\end{aligned} \tag{36}$$

For the radial component, we find:

$$B_r(r, z) = \sum_{n=1}^{\infty} \frac{(-1)^n}{n!(n-1)!} \frac{\partial^{2n-1} B(z)}{\partial z^{2n-1}} \left(\frac{r}{2} \right)^{2n-1}. \tag{37}$$

A similar procedure gives the radial and longitudinal components of the electric field. Define $V(z)$ as the on-axis value of the electric potential ϕ :

$$V(z) \equiv \phi(0, z). \quad (38)$$

This gives:

$$\begin{aligned} E_z(r, z) &= -\frac{\partial \phi}{\partial z} = -\sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} \frac{\partial^{2n+1} V(z)}{\partial z^{2n+1}} \left(\frac{r}{2}\right)^{2n}. \\ E_r(r, z) &= -\frac{\partial \phi}{\partial r} = -\sum_{n=1}^{\infty} \frac{(-1)^n}{n!(n-1)!} \frac{\partial^{2n} V(z)}{\partial z^{2n}} \left(\frac{r}{2}\right)^{2n-1}. \end{aligned} \quad (39)$$

Derivation of paraxial ray equation

We will now take the *paraxial* limit, assuming small transverse position and velocity: $\dot{r} \ll \dot{z}$, $r\dot{\theta} \ll \dot{z}$. If $v = (\dot{r}^2 + r^2\dot{\theta}^2 + \dot{z}^2)^{1/2}$ is the total velocity, the paraxial limit gives $\dot{z} = (v^2 - \dot{r}^2 - r^2\dot{\theta}^2)^{1/2} \approx v$.

To first order in r and r' , the external fields are:

$$\begin{aligned} E_r^{(ext)} &\approx \frac{1}{2} \frac{\partial^2 V(z)}{\partial z^2} r, \\ B_z^{(ext)} &\approx B(z), \\ B_\theta^{(ext)} &= 0 \quad (\text{since } \partial\phi_m/\partial\theta = 0). \end{aligned} \tag{40}$$

Plug these truncated fields into the radial equation of motion (Eq. (12)):

$$\begin{aligned} \frac{d(\gamma m \dot{r})}{dt} - \gamma m r \dot{\theta}^2 &= q \left(E_r + r \dot{\theta} B_z - \dot{z} B_\theta \right), \\ &\approx q \left(\frac{1}{2} \frac{\partial^2 V(z)}{\partial z^2} r + r \dot{\theta} B(z) \right) \end{aligned} \tag{41}$$

Use z as the independent variable: $dz = \beta c dt$. Using $' \equiv d/dz$ and dropping the $\approx$ symbol gives:

$$\beta c \frac{d(\gamma m \beta c r')}{dz} - \gamma m r (\beta c \theta')^2 = q \left(\frac{1}{2} V''(z) r + r \beta c \theta' B(z) \right). \quad (42)$$

Expand the derivative in the first term and divide by $\gamma \beta^2 m c^2$:

$$r'' + \frac{(\gamma \beta)'}{(\gamma \beta)} r' - r \theta'^2 = \frac{q}{\gamma \beta^2 m c^2} \left(\frac{1}{2} V''(z) + \beta c \theta' B(z) \right) r. \quad (43)$$

Now use the definition of the canonical angular momentum p_θ (Eq. (21)) to eliminate θ' , noting that the flux $\psi(r) = \pi r^2 B$:

$$\begin{aligned}
 \theta' &= \frac{\dot{\theta}}{\beta c} \\
 &= \frac{1}{\beta c} \frac{1}{\gamma m r^2} \left(p_\theta - \frac{q\psi(r)}{2\pi} \right), \\
 &= \frac{p_\theta}{\gamma \beta m c r^2} - \frac{\omega_c}{2\gamma \beta c}.
 \end{aligned} \tag{44}$$

We have defined the *cyclotron frequency* $\omega_c \equiv \frac{qB}{m}$.

Substitute Eq. (44) into Eq. (43):

$$r'' + \frac{(\gamma\beta)'}{\gamma\beta} r + \left(\frac{\omega_c}{2\gamma\beta c} \right)^2 r - \left(\frac{p_\theta}{\gamma\beta m c} \right)^2 \frac{1}{r^3} = \frac{q}{\gamma m \beta^2 c^2} \frac{1}{2} V''(z) r \tag{45}$$

The rate of change in the particle energy is given by

$$\frac{d(\gamma mc^2)}{dt} = q\mathbf{E} \cdot \mathbf{v}. \quad (46)$$

Rearrange and switch from t to z . In the paraxial limit:

$$\gamma' = \frac{q}{mc^2} \frac{\mathbf{E} \cdot \mathbf{v}}{v_z} \quad (47)$$

$$= \frac{q}{mc^2} \frac{E_x v_x + E_y v_y + E_z v_z}{v_z} \quad (48)$$

$$\approx \frac{q}{mc^2} E_z. \quad (49)$$

Giving the approximate relationship:

$$\gamma'' \approx \frac{q}{mc^2} E'_z = -\frac{q}{mc^2} V''. \quad (50)$$

This approximation of λ'' leads to the *paraxial ray equation*:

$$\boxed{r'' + \frac{(\gamma\beta)'}{(\gamma\beta)}r' + \left(\frac{\gamma''}{2\gamma\beta^2}\right)r + \left(\frac{\omega_c}{2\gamma\beta c}\right)^2 r - \left(\frac{p_\theta}{\gamma\beta mc}\right)^2 \frac{1}{r^3} = 0.} \quad (51)$$

Terms in order: (i) inertial term; (ii) accelerative damping; (iii) E_r from converging field lines; (iv) solenoidal focusing ($v_\theta B_z$, part of centrifugal term); (v) centrifugal defocusing.

Addition of self-generated fields

We can also account for linearized self-fields $\mathbf{E}^{(self)}$ and $\mathbf{B}^{(self)}$. Adding these self-generated fields to the radial equation of motion gives:

$$r'' + \frac{(\gamma\beta)'}{(\gamma\beta)} - r\theta'^2 = \frac{q}{mc^2\gamma\beta^2} \left(\frac{1}{2}V''(z)r + \beta c\theta' B(z)r + E_r^{(self)} - \beta c B_\theta^{(self)} \right). \quad (52)$$

So we only need the r component of the self-generated electric field and the θ component of the self-generated magnetic field. We will ignore the $B_z^{(self)}$ term in the paraxial limit.

Recall that $V(z) \equiv \phi(0, z)$ is the external potential ϕ on-axis, and to linear order $E_r(r, z) \approx V''(z)r/2$. Similarly, $B(z) \equiv B_z(0, z)$ is the external magnetic field on-axis, and to linear order $B_z(r, z) \approx B(z)$.

Assuming cylindrical symmetry of the beam, the Poisson equation for the self-generated electric potential is:

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi^{(self)}}{\partial r} \right) + \frac{\partial^2 \phi^{(self)}}{\partial s^2} = -\frac{\rho(r)}{\epsilon_0}. \quad (53)$$

Solving for $\partial \phi^{(self)} / \partial r$ relates the radial electric field to z -derivative of the potential:

$$E_r^{(self)} = -\frac{\partial \phi^{(self)}}{\partial r} = \frac{\lambda(r)}{2\pi\epsilon_0 r} + \frac{1}{2} \frac{\partial^2 \phi^{(self)}}{\partial z^2} r, \quad (54)$$

where λ is the line-charge density,

$$\lambda(r) \equiv \int_0^r 2\pi\rho(\tilde{r})\tilde{r}d\tilde{r}. \quad (55)$$

Above, it is assumed that $\partial^2 \phi^{(self)} / \partial z^2$ has no dependence on r (keep lowest-order term).

For the θ component of the self-generated magnetic field, use Ampere's law:

$$\oint \mathbf{B}^{(self)} \cdot d\mathbf{l} = \mu_0 I_{enc} \quad (56)$$

$$2\pi r B_{\theta}^{(self)} = \mu_0 \int_0^r 2\pi \tilde{r} J_z(\tilde{r}) d\tilde{r} = \mu_0 \lambda(r) v_z \quad (57)$$

$$B_{\theta}^{(self)} = \frac{\mu_0 \beta c \lambda(r)}{2\pi r} = \frac{\beta}{c} \frac{\lambda(r)}{2\pi \epsilon_0 r}. \quad (58)$$

We can now evaluate $E_r^{(self)} - \beta c B_\theta^{(self)}$:

$$E_r^{(self)} - \beta c B_\theta^{(self)} = \frac{1}{\gamma^2} \frac{\lambda(r)}{2\pi\epsilon_0 r} + \frac{1}{2} \frac{\partial^2 \phi^{(self)}}{\partial z^2} r, \quad (59)$$

We can also repeat the approximation of γ'' with the inclusion of the self-generated electric field.

$$\gamma'' \approx -\frac{q}{mc^2} \frac{\partial^2}{\partial z^2} (V + \phi^{(self)}). \quad (60)$$

The result is a driving term on the right side of the paraxial ray equation:

$$\boxed{r'' + \frac{(\gamma\beta)'}{(\gamma\beta)} r' + \left(\frac{\gamma''}{2\gamma\beta^2} \right) r + \left(\frac{\omega_c}{2\gamma\beta c} \right)^2 r - \left(\frac{p_\theta}{\gamma\beta mc} \right)^2 \frac{1}{r^3} = \frac{q}{mc^2 \beta^2 \gamma^3} \frac{\lambda(r)}{2\pi\epsilon_0 r}} \quad (61)$$

Envelope equations for axially symmetric beams

Time-evolution of expected value

Let $g(\mathbf{x}, \mathbf{x}')$ be any function of the phase space coordinates. Use brackets to indicate the expected value of g under probability distribution f :

$$\langle g \rangle \equiv \iint g(\mathbf{x}, \mathbf{x}') f(\mathbf{x}, \mathbf{x}', s) d\mathbf{x} d\mathbf{x}'. \quad (62)$$

What is the time-evolution of $\langle g \rangle$?

$$\frac{d\langle g \rangle}{ds} = \frac{d}{ds} \iint g(\mathbf{x}, \mathbf{x}') f(\mathbf{x}, \mathbf{x}', s) d\mathbf{x} d\mathbf{x}' \quad (63)$$

$$\begin{aligned}
\frac{d\langle g \rangle}{ds} &= \frac{d}{ds} \iint g(\mathbf{x}, \mathbf{x}') f(\mathbf{x}, \mathbf{x}') d\mathbf{x} d\mathbf{x}' \\
&= \iint g(\mathbf{x}, \mathbf{x}') \frac{\partial f(\mathbf{x}, \mathbf{x}', s)}{\partial s} d\mathbf{x} d\mathbf{x}' \quad (\text{Leibniz Rule}) \\
&= \iint g(\mathbf{x}, \mathbf{x}') \left[-\mathbf{x}' \cdot \frac{\partial f}{\partial \mathbf{x}} - \mathbf{x}'' \cdot \frac{\partial f}{\partial \mathbf{x}'} \right] d\mathbf{x} d\mathbf{x}' \quad (\text{Vlasov equation}).
\end{aligned} \tag{64}$$

Since f vanishes at the integration boundary (∞), we can use integration by parts to flip the gradients from f to g :

$$\begin{aligned}\iint \left[\mathbf{x}' \cdot \frac{\partial f}{\partial \mathbf{x}} g \right] d\mathbf{x} d\mathbf{x}' &= - \iint \left[\mathbf{x}' \cdot \frac{\partial g}{\partial \mathbf{x}} f \right] d\mathbf{x} d\mathbf{x}' \\ &= -\langle \mathbf{x}' \cdot \partial g / \partial \mathbf{x} \rangle.\end{aligned}\tag{65}$$

$$\begin{aligned}\iint \left[\mathbf{x}'' \cdot \frac{\partial f}{\partial \mathbf{x}'} g \right] d\mathbf{x} d\mathbf{x}' &= - \iint \left[\mathbf{x}'' \cdot \frac{\partial g}{\partial \mathbf{x}'} f \right] d\mathbf{x} d\mathbf{x}' \\ &= -\langle \mathbf{x}'' \cdot \partial g / \partial \mathbf{x}' \rangle.\end{aligned}\tag{66}$$

Integration by parts (one-dimensional):

$$\int_a^b f(x)g'(x)dx = [f(x)g(x)]_a^b - \int_a^b f'(x)g(x)dx. \quad (67)$$

$$\begin{aligned}
& \cdot = \iint g(\mathbf{x}, \mathbf{x}') \mathbf{x}' \cdot \frac{\partial f(\mathbf{x}, \mathbf{x}', s)}{\partial \mathbf{x}} d\mathbf{x} d\mathbf{x}' \\
& = \iiint g(x, x', y, y') \left[x' \frac{\partial}{\partial x} + y' \frac{\partial}{\partial y} \right] f(x, x', y, y', s) dx dx' dy dy' \\
& = \iiint x' \left[\int g \frac{\partial f}{\partial x} dx \right] dx' dy dy' + \iiint y' \left[\int g \frac{\partial f}{\partial y} dy \right] dx dx' dy' \\
& = - \iiint f \left[x' \frac{\partial g}{\partial x} + x' \frac{\partial g}{\partial x} \right] dx dx' dy dy' \\
& = - \iint f(\mathbf{x}, \mathbf{x}', s) \left[\mathbf{x}' \cdot \frac{\partial g(\mathbf{x}, \mathbf{x}')}{\partial \mathbf{x}} \right] d\mathbf{x} d\mathbf{x}' \\
& = - \langle \mathbf{x}' \cdot \frac{\partial g}{\partial \mathbf{x}} \rangle.
\end{aligned} \tag{68}$$

We can then write $d\langle g \rangle/ds$ as:

$$\frac{d\langle g \rangle}{ds} = \langle \mathbf{x}' \cdot \frac{\partial g}{\partial \mathbf{x}} \rangle + \langle \mathbf{x}'' \cdot \frac{\partial g}{\partial \mathbf{x}'} \rangle. \quad (69)$$

From the definition of the total derivative:

$$\frac{dg}{ds} = \mathbf{x}' \cdot \frac{\partial g}{\partial \mathbf{x}} + \mathbf{x}'' \cdot \frac{\partial g}{\partial \mathbf{x}'}. \quad (70)$$

Therefore, the time-derivative of the expected value is the same as the expected value of the time-derivative:

$$\frac{d\langle g \rangle}{ds} = \left\langle \frac{dg}{ds} \right\rangle. \quad (71)$$

To evaluate $d\langle g \rangle/ds$, we can evaluate dg/ds and then perform the averaging. For example:

$$\frac{d}{ds}\langle xx \rangle = 2\langle xx' \rangle \tag{72}$$

$$\frac{d}{ds}\langle x'x' \rangle = 2\langle x'x'' \rangle \tag{73}$$

$$\frac{d}{ds}\langle xx' \rangle = \langle xx'' \rangle + \langle x'x' \rangle \tag{74}$$

$$\vdots \tag{75}$$

Evolution of second moments

Define the root-mean-square (rms) radius $\tilde{r} \equiv \sqrt{\langle r^2 \rangle}$.

Evaluate $\tilde{r}'$:

$$\begin{aligned}\frac{d}{ds} \tilde{r}^2 &= \frac{d}{ds} \langle r^2 \rangle \\ \tilde{r}' &= \frac{\langle rr' \rangle}{\tilde{r}}.\end{aligned}\tag{76}$$

Evaluate $\tilde{r}''$:

$$\begin{aligned}\tilde{r}'' &= \frac{\langle rr'' \rangle + \langle r' r' \rangle}{\tilde{r}} - \frac{\langle rr' \rangle \tilde{r}'}{\tilde{r}^2} \\ \tilde{r}'' &= \frac{\langle rr'' \rangle}{\tilde{r}} + \frac{\langle rr \rangle \langle r' r' \rangle - \langle rr' \rangle \langle rr' \rangle}{\tilde{r}^3}.\end{aligned}\tag{77}$$

What is $\langle rr'' \rangle$?

Evaluation of $\langle rr'' \rangle$

Return to the paraxial ray equation for the radius r (including linear self-fields):

$$r'' + \frac{(\gamma\beta)'}{(\gamma\beta)} r' + \left(\frac{\gamma''}{2\gamma\beta^2} \right) r + \left(\frac{\omega_c}{2\gamma\beta c} \right)^2 r - \left(\frac{p_\theta}{\gamma\beta mc} \right)^2 \frac{1}{r^3} - \frac{q}{mc^2\beta^2\gamma^3} \frac{\lambda(r)}{2\pi\epsilon_0 r} = 0. \quad (78)$$

Rewrite as:

$$r'' + \frac{(\gamma\beta)'}{(\gamma\beta)} r' + \left(\frac{\gamma''}{2\gamma\beta^2} \right) r - \frac{\omega_c \theta' r}{\gamma\beta c} - \theta'^2 r - \frac{q}{mc^2\beta^2\gamma^3} \frac{\lambda(r)}{2\pi\epsilon_0 r} = 0. \quad (79)$$

Multiply by r and take averages:

$$\langle rr'' \rangle + \frac{(\gamma\beta)'}{(\gamma\beta)} \langle rr' \rangle + \left(\frac{\gamma''}{2\gamma\beta^2} \right) \langle r^2 \rangle - \frac{\omega_c}{\gamma\beta c} \langle \theta' r^2 \rangle - \langle \theta'^2 r^2 \rangle - \frac{q}{mc^2\beta^2\gamma^3} \frac{\langle \lambda(r) \rangle}{2\pi\epsilon_0} = 0. \quad (80)$$

$$p_{\theta} \equiv \gamma\beta mcr^2\theta' + \frac{m\omega_c r^2}{2} \quad (81)$$

$$\langle p_{\theta} \rangle = \gamma\beta mc \langle r^2 \theta' \rangle + \frac{m\omega_c \langle r^2 \rangle}{2} \quad (82)$$

$$\vdots \quad (83)$$

$$\frac{\omega_c}{\gamma\beta c} \langle r^2 \theta' \rangle = \left(\frac{1}{\gamma\beta mc} \right)^2 \frac{\langle p_{\theta} \rangle^2}{\tilde{r}^2} - \frac{\langle r^2 \theta' \rangle^2}{\tilde{r}^2} - \left(\frac{\omega_c}{2\gamma\beta c} \right)^2 \tilde{r}^2 \quad (84)$$

$$\boxed{\tilde{r}'' + \frac{(\gamma\beta)'}{(\gamma\beta)}\tilde{r}' + \left(\frac{\gamma''}{2\gamma\beta^2}\right)\tilde{r} + \left(\frac{\omega_c}{2\gamma\beta c}\right)^2\tilde{r} - \left(\frac{\langle p_\theta \rangle}{\gamma\beta mc}\right)^2\frac{1}{\tilde{r}^3} - \frac{\varepsilon_r^2}{\tilde{r}^3} - \frac{Q}{\tilde{r}} = 0}$$

(85)

$$Q \equiv \frac{1}{mc^2\beta^2\gamma^3} \frac{q\langle\lambda(r)\rangle}{2\pi\epsilon_0}$$

(86)

$$\varepsilon_r \equiv \sqrt{\langle r^2 \rangle \langle r'^2 \rangle - \langle rr' \rangle^2 + \langle r^2 \rangle \langle r^2 \theta'^2 \rangle - \langle r^2 \theta' \rangle^2}$$

(87)

Envelope equations for elliptically symmetric beams

Equations of motion ($\mathbf{x} = [x, y]$):

$$\frac{d(\gamma m \dot{x})}{dt} = q (E_x + \dot{y} B_z - \dot{z} B_y) \quad (88)$$

$$\frac{d(\gamma m \dot{y})}{dt} = q (E_y + \dot{z} B_x - \dot{x} B_z) \quad (89)$$

Use z as independent variable: $dz = \beta c dt$:

$$x'' + \frac{(\gamma\beta)'}{(\gamma\beta)} x' = \frac{q}{\gamma m \beta^2 c^2} [E_x + \beta c (y' B_z - B_y)] \quad (90)$$

$$y'' + \frac{(\gamma\beta)'}{(\gamma\beta)} y' = \frac{q}{\gamma m \beta^2 c^2} [E_y + \beta c (B_x - x' B_z)] \quad (91)$$

Add electrostatic potential ϕ in beam rest frame:

$$x'' + \frac{(\gamma\beta)'}{(\gamma\beta)} x' = \frac{q}{\gamma m \beta^2 c^2} \left[E_x + \beta c (y' B_z - B_y) - \frac{1}{\gamma^2} \frac{\partial^2 \phi}{\partial x} \right] \quad (92)$$

$$y'' + \frac{(\gamma\beta)'}{(\gamma\beta)} y' = \frac{q}{\gamma m \beta^2 c^2} \left[E_y + \beta c (B_x - x' B_z) - \frac{1}{\gamma^2} \frac{\partial^2 \phi}{\partial y} \right] \quad (93)$$

Assume no external electric field ($E_x = E_y = 0$) and approximate magnetic field as $B_y = Gx$, $B_x = -Gy$.

[...]

$$\tilde{x} = \sqrt{\langle xx \rangle} \tag{94}$$

$$\tilde{x}' = \frac{\langle xx' \rangle}{\tilde{x}} \tag{95}$$

$$\tilde{x}'' = \frac{\langle xx'' \rangle}{\tilde{x}} + \frac{\langle xx \rangle \langle x'x' \rangle - \langle xx' \rangle \langle xx' \rangle}{\tilde{x}^3} \tag{96}$$

$$= \frac{\langle xx'' \rangle}{\tilde{x}} + \frac{\varepsilon_x^2}{\tilde{x}^3} \tag{97}$$

